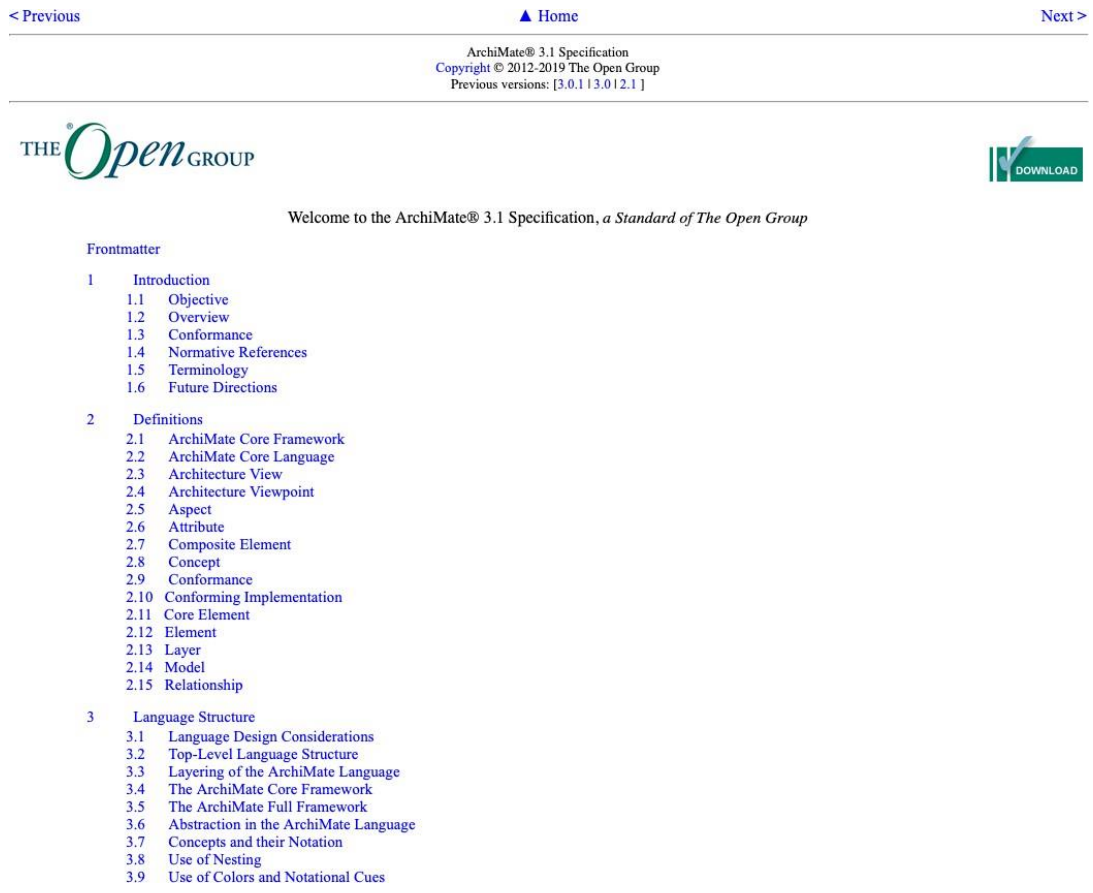
Archimate
Technology Layer

ArchiMate Documentation

https://pubs.opengroup.org/architecture/archimate3-doc/ch-Technology-Layer.html



ArchiMate® 3.1 Specification
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Previous versions: [3.0.1 | 3.0 | 2.1]

THE Open GROUP

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Welcome to the ArchiMate® 3.1 Specification, *a Standard of The Open Group*

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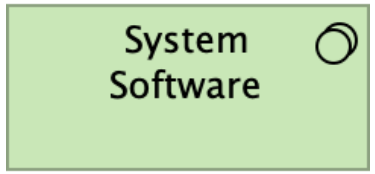
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Active Elements



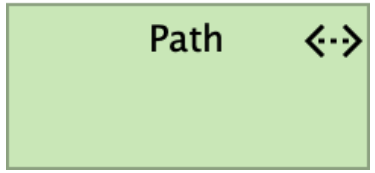
A physical IT asset on which software can be run and/or artifacts (artifacts) can be found



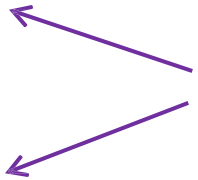
System software that provides basic functionality (store, execute) useful for building applications



Communication structure to connect two or more devices

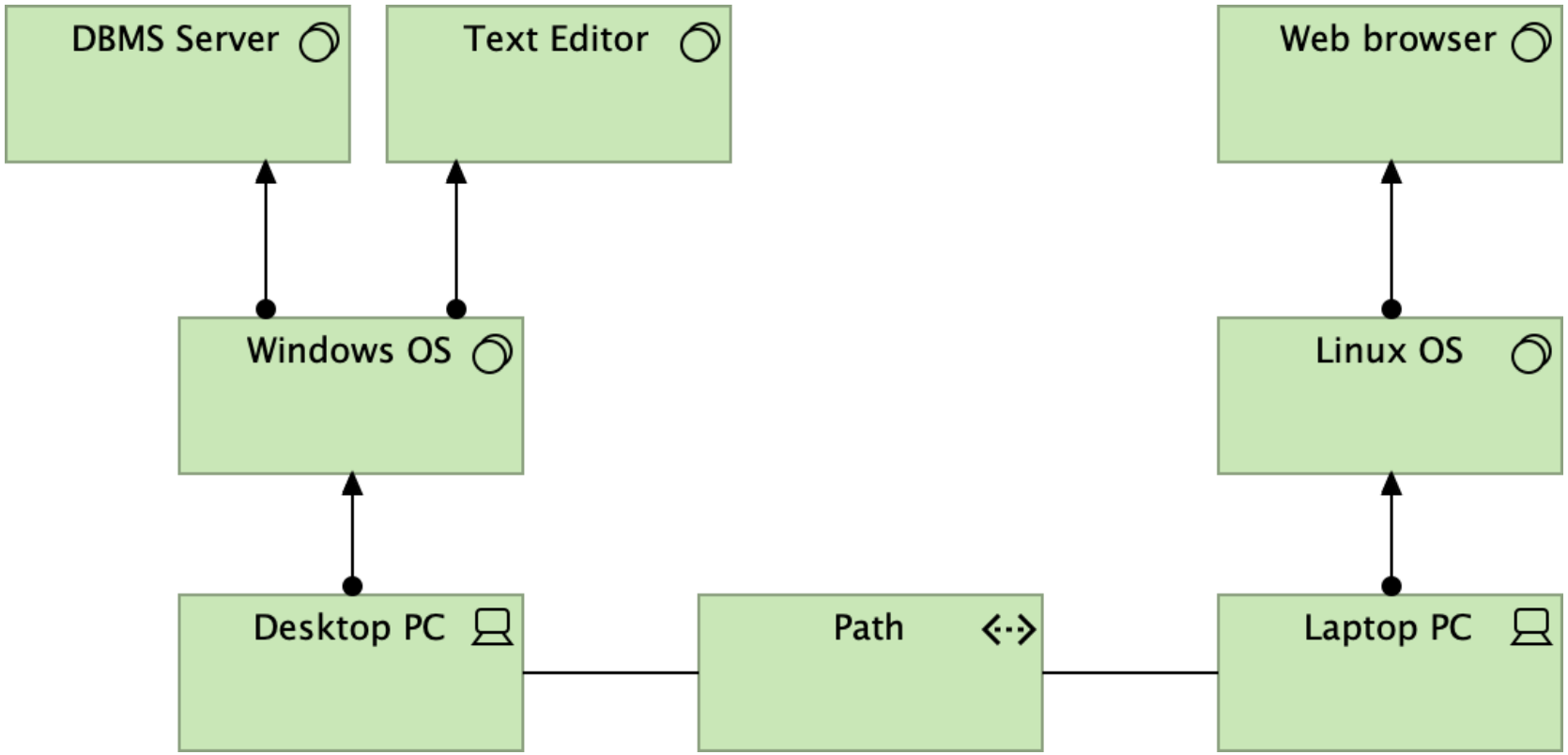


Abstraction of end-to-end communication: it is possible to identify a connection between nodes (without going into the technological detail of the implementation)

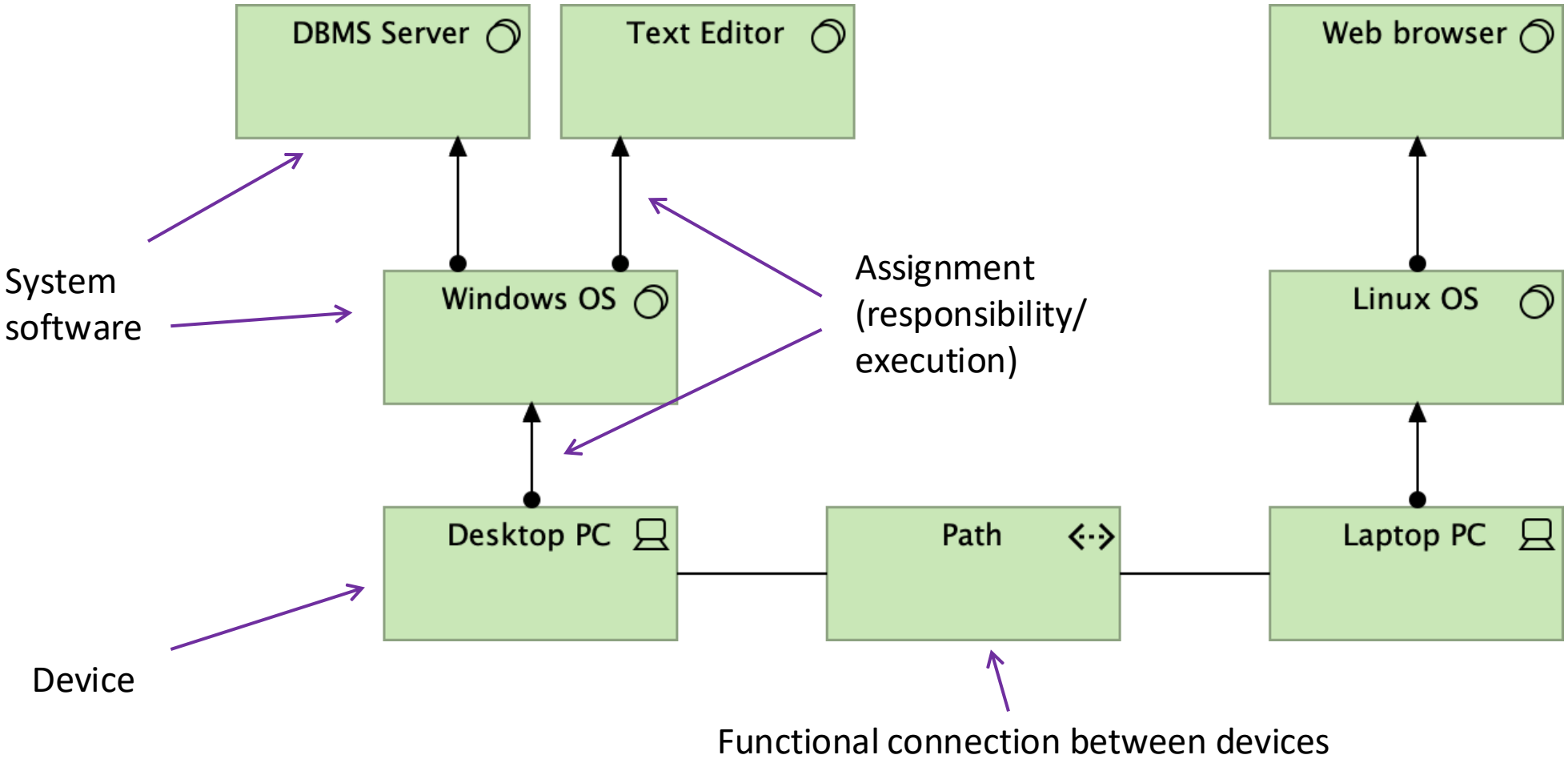


Generic Association (solid line)

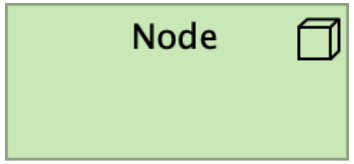
Active Elements: example



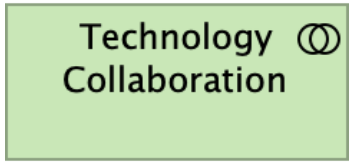
Active Elements: example



Elementi attivi



A physical or computational resource (including compound) relevant to our IT infrastructure that hosts, manipulates, or interacts with other resources

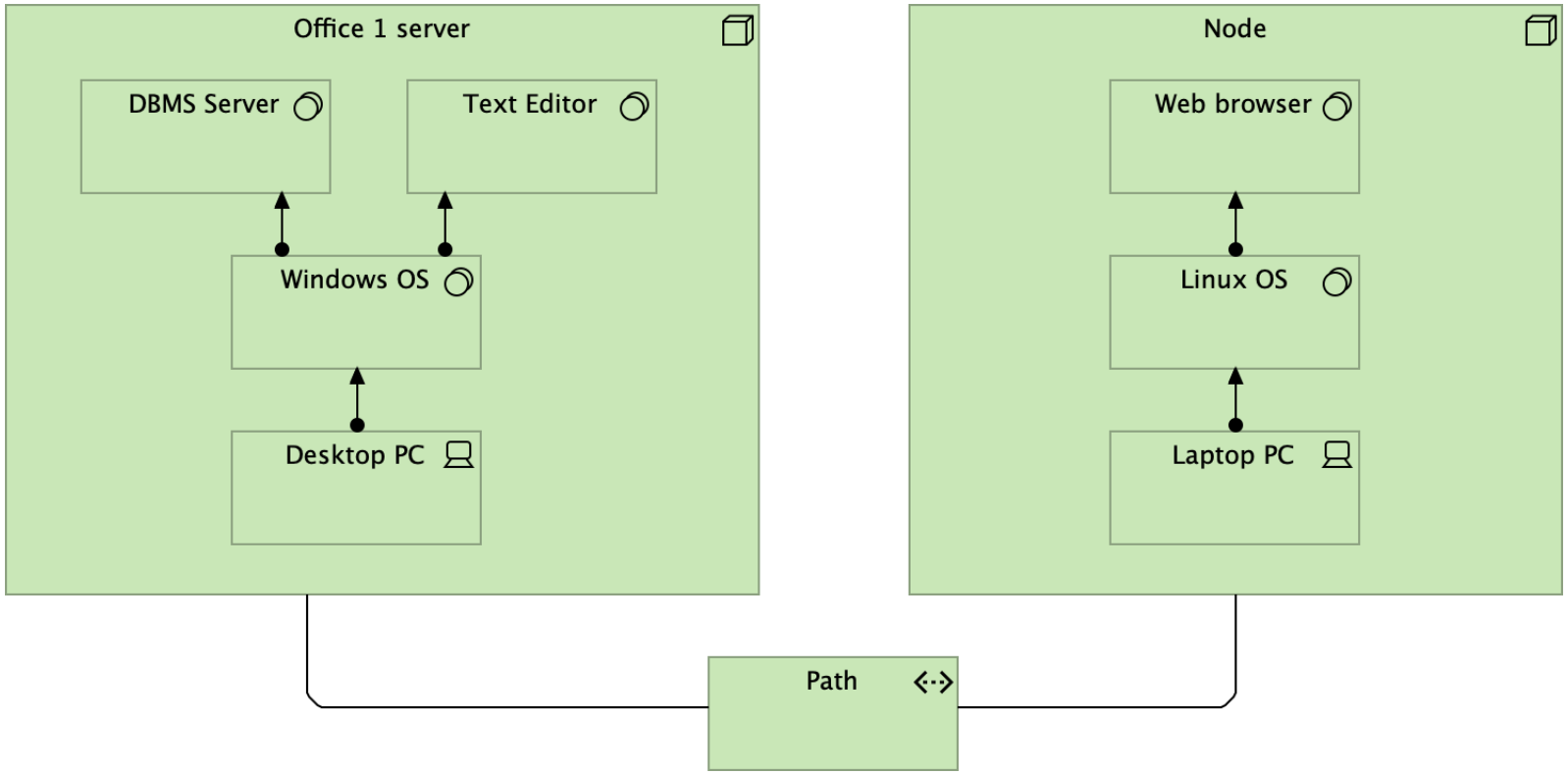


An aggregate of two or more nodes working together to achieve a certain goal



The access point that allows you to use the functionality made available by a node

Active Elements: Example



Nodes can be used to logically group elements. Typically, elements linked by assignment are aggregated in a node

Behavioral elements



A sequence of technological behaviors that achieve a specific result



A group of infrastructure behaviors that can be run by a node



A collective behavior performed by one or more nodes



An externally visible functional unit, provided by one or more nodes, exposed through well-defined interfaces, performing a specific task. It can be used by other elements at the technological level, or at the next level



A change of state at the technological level

Behavioral elements

The complexity at the technological level is greater than in the business and application layers

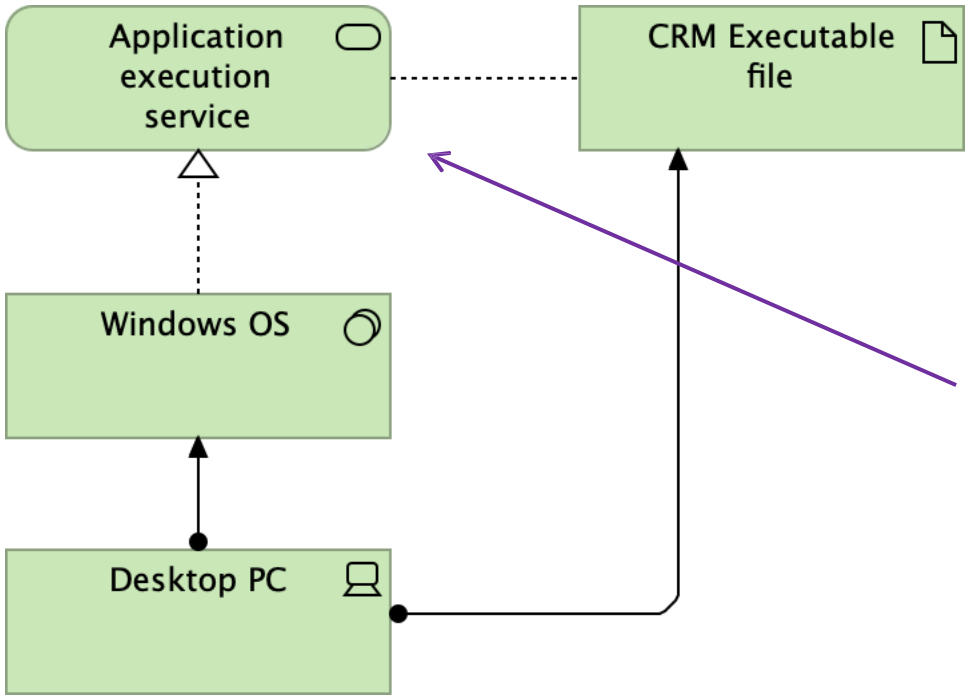
In the technology layer there is no single "pattern base" but we will work on typical examples of IT infrastructure deployments to be used as reference points (tier/layer applications)

Passive elements



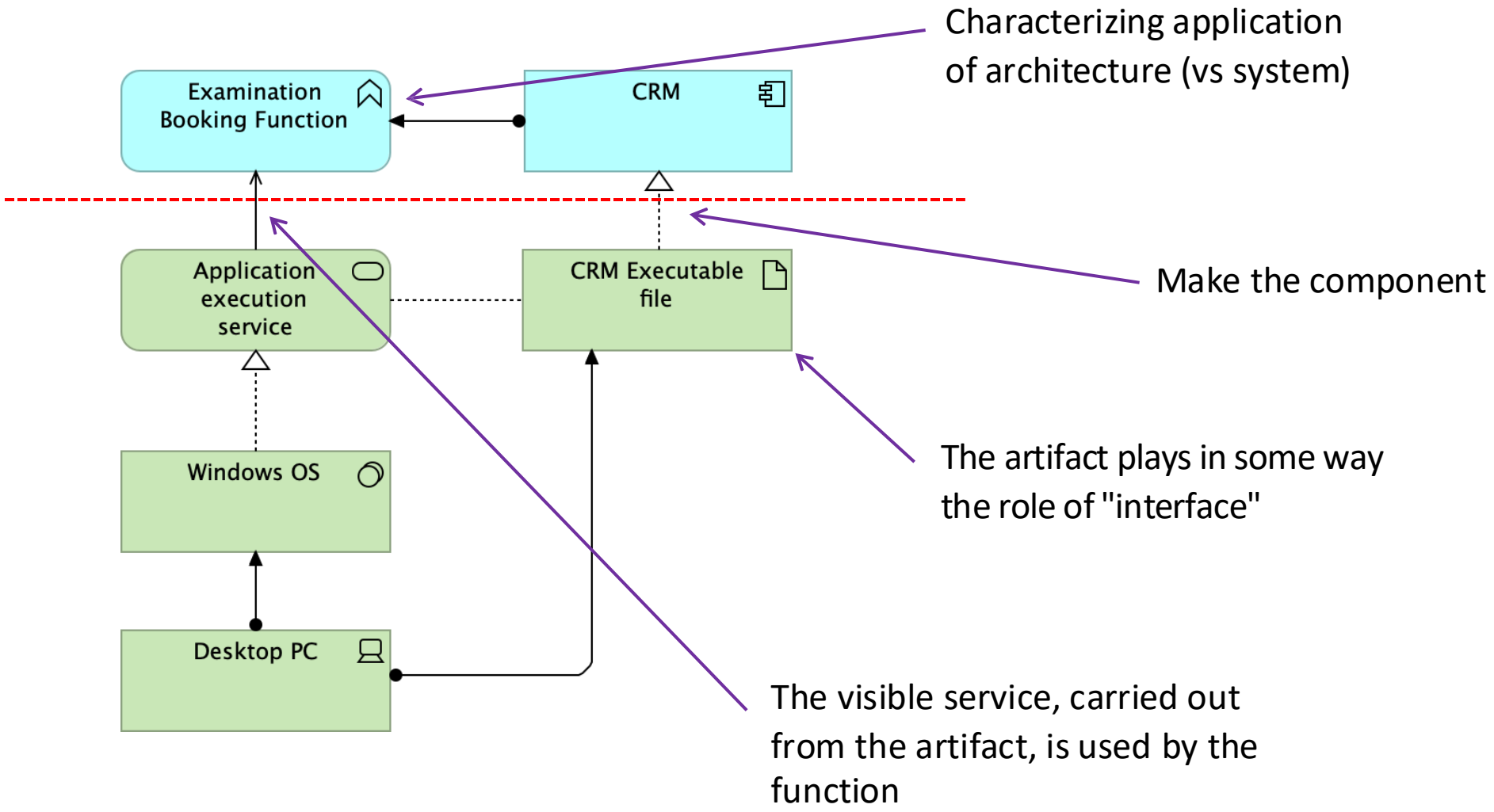
A piece of data or fragment (e.g. file) that can be produced or used within an IT infrastructure (we are at the "physical" level of architecture)

Software system vs application

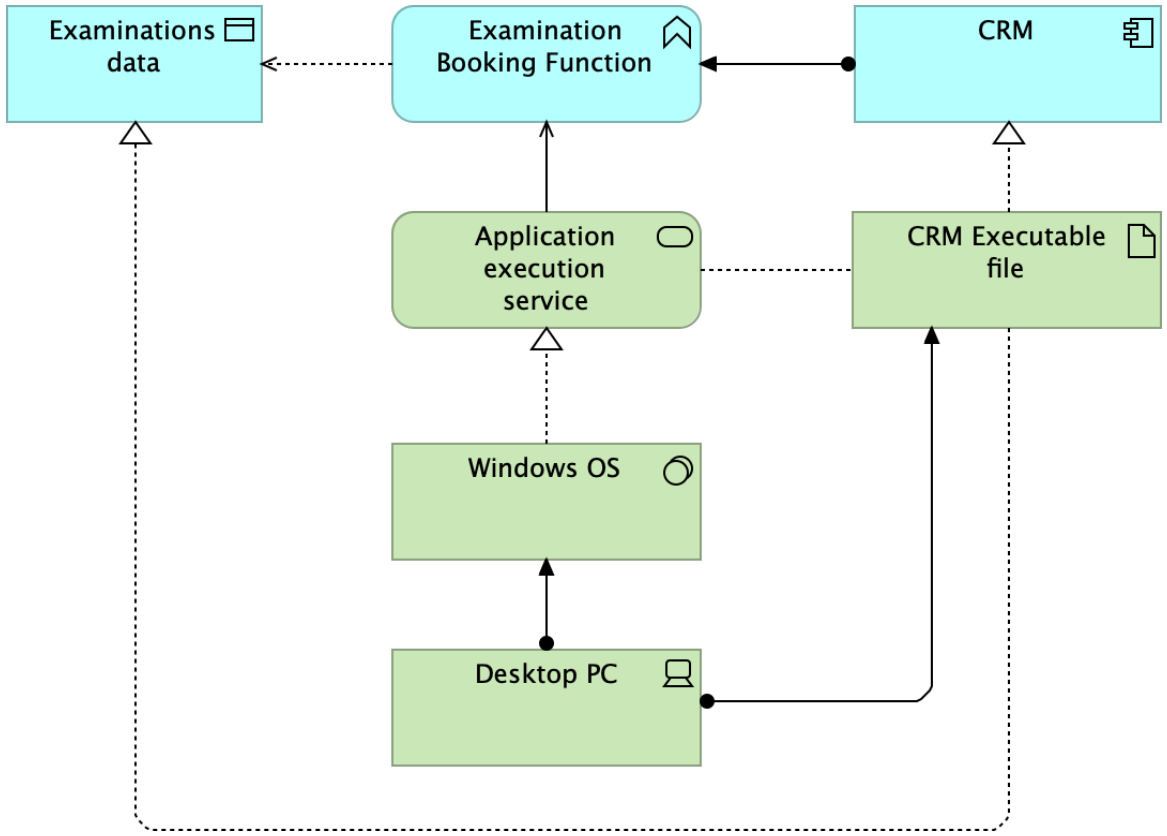


OS allows me to run a file for exposing a certain service (execution of the artifact that makes a service available)

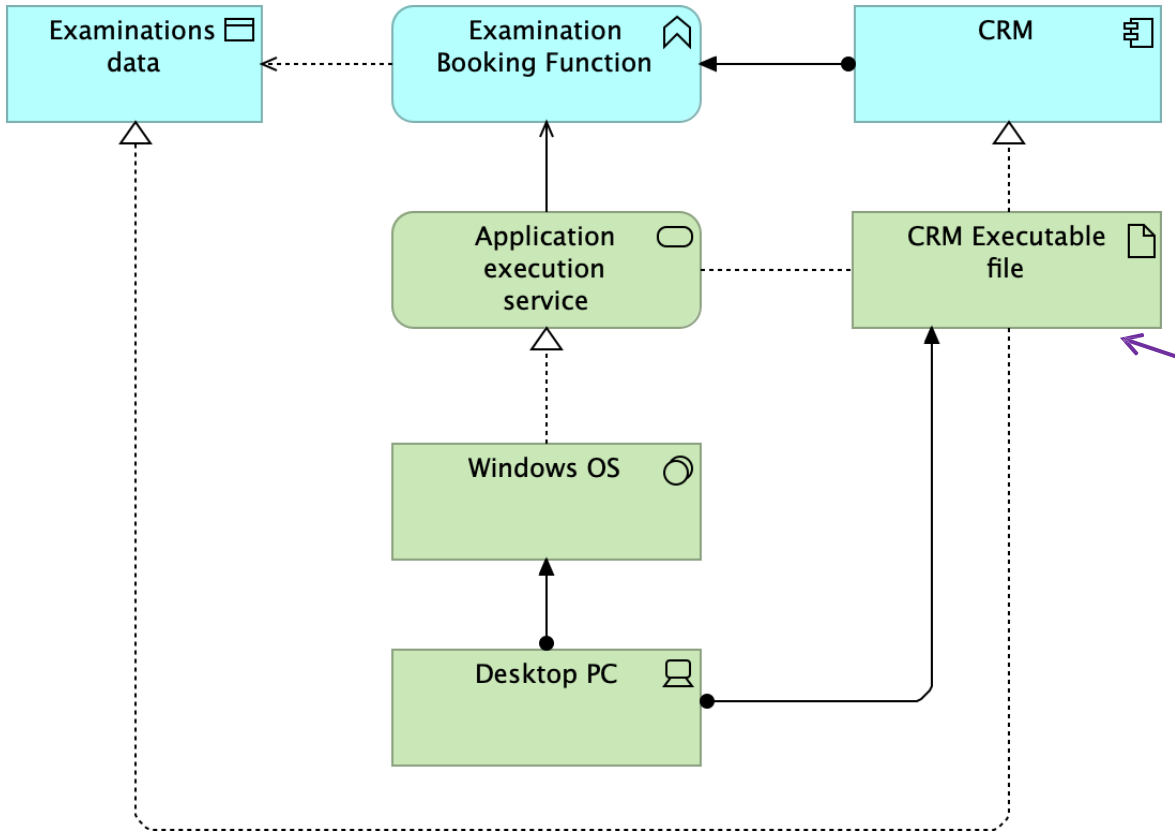
System Software vs Application



Standalone application



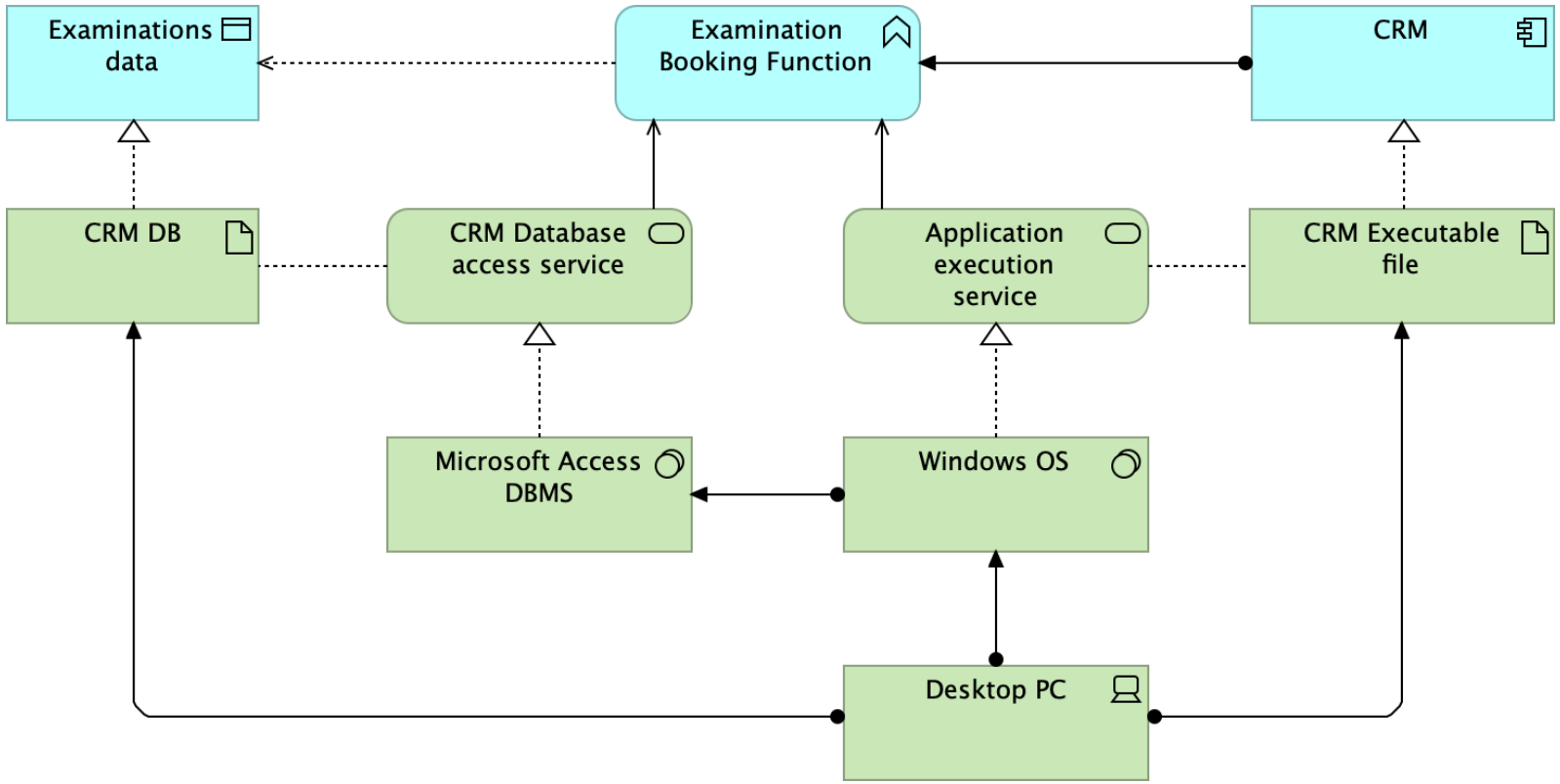
Standalone application



An application contained in a single executable file, which also includes data

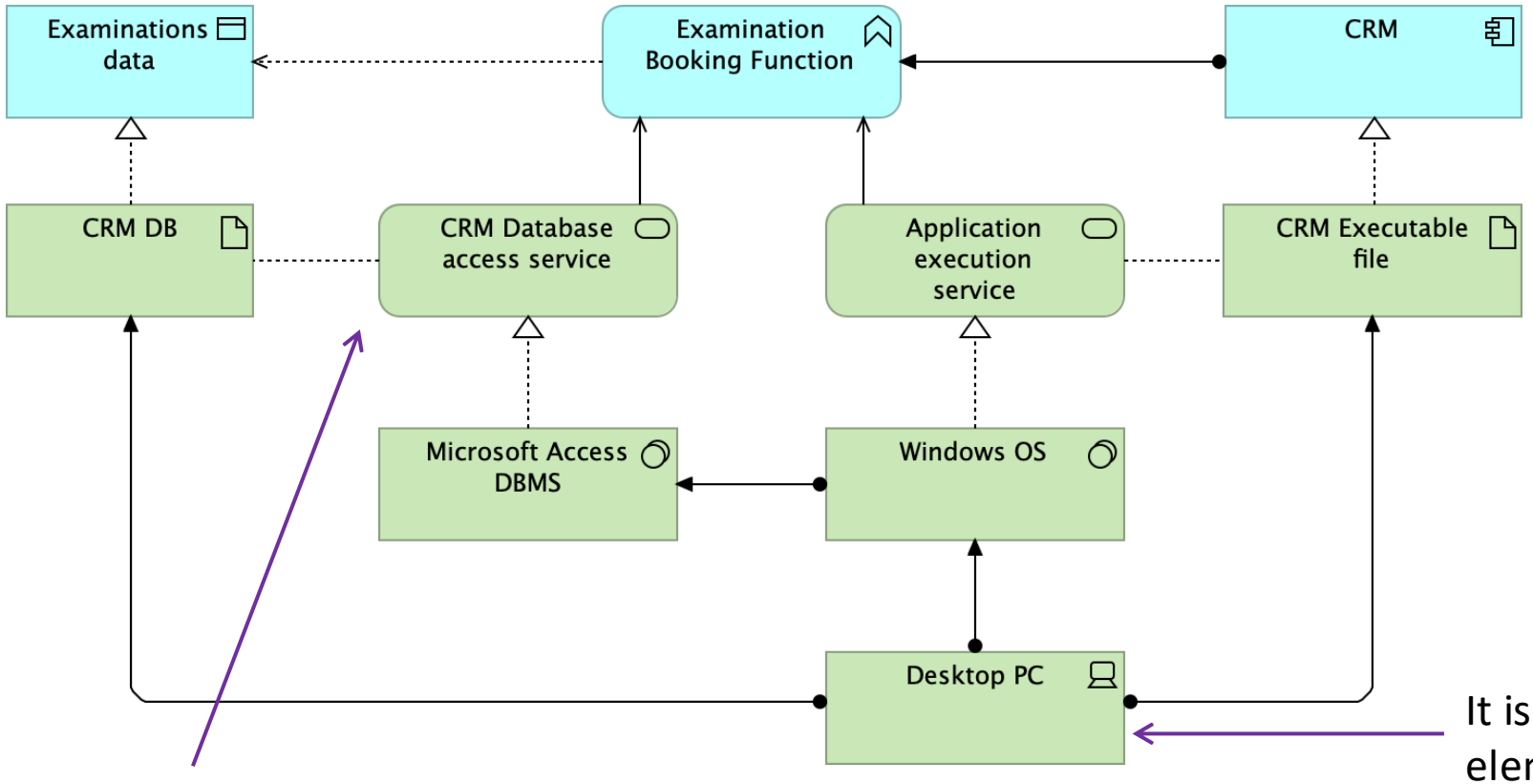
(NOT REAL! How to describe a more realistic situation?)

One-tiered application



The application is deployed within a single physical element

One-tiered application

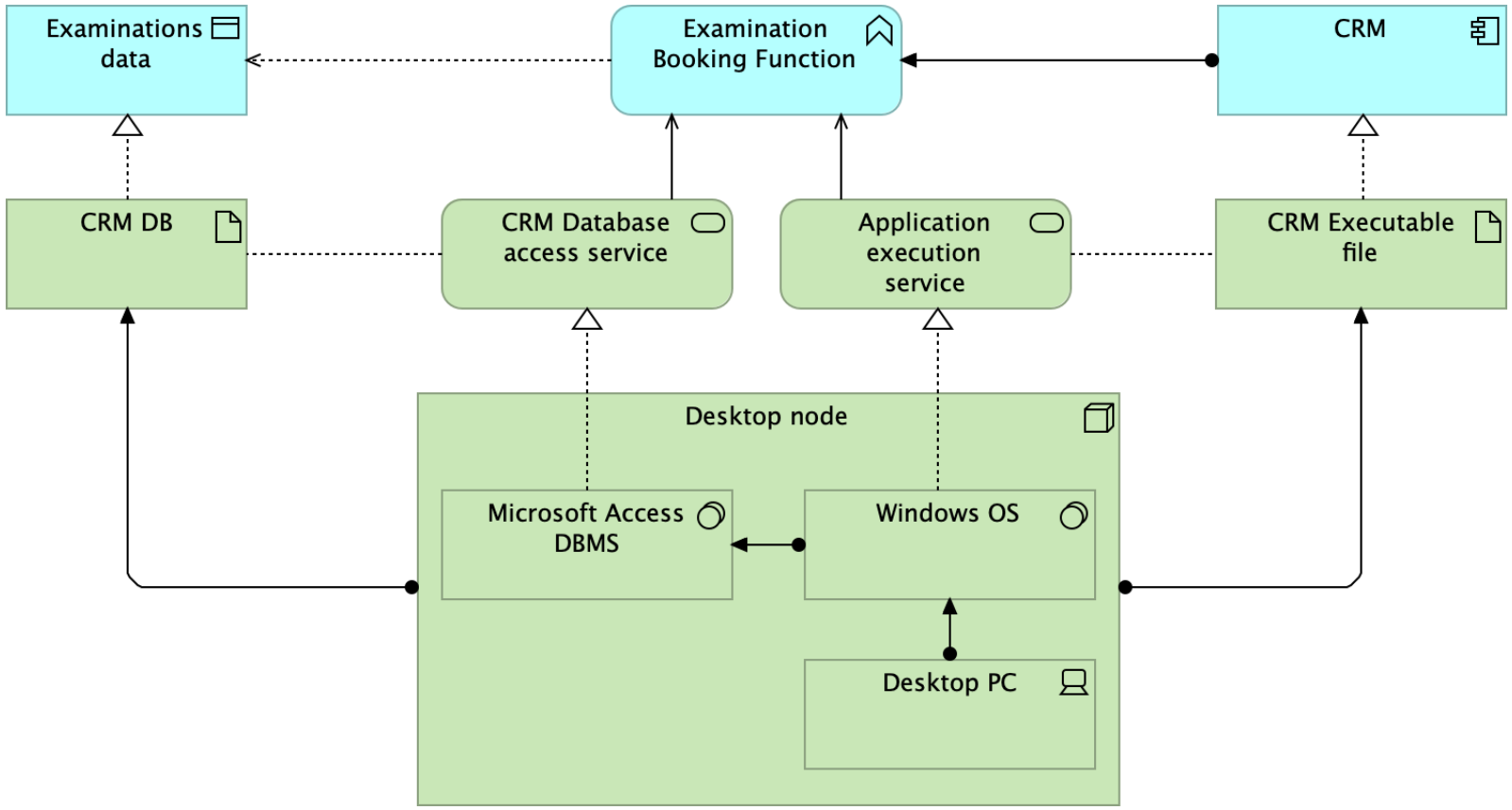


Section that contains all application and presentation logic

Section that concerns the data management logic, managed by a DBMS

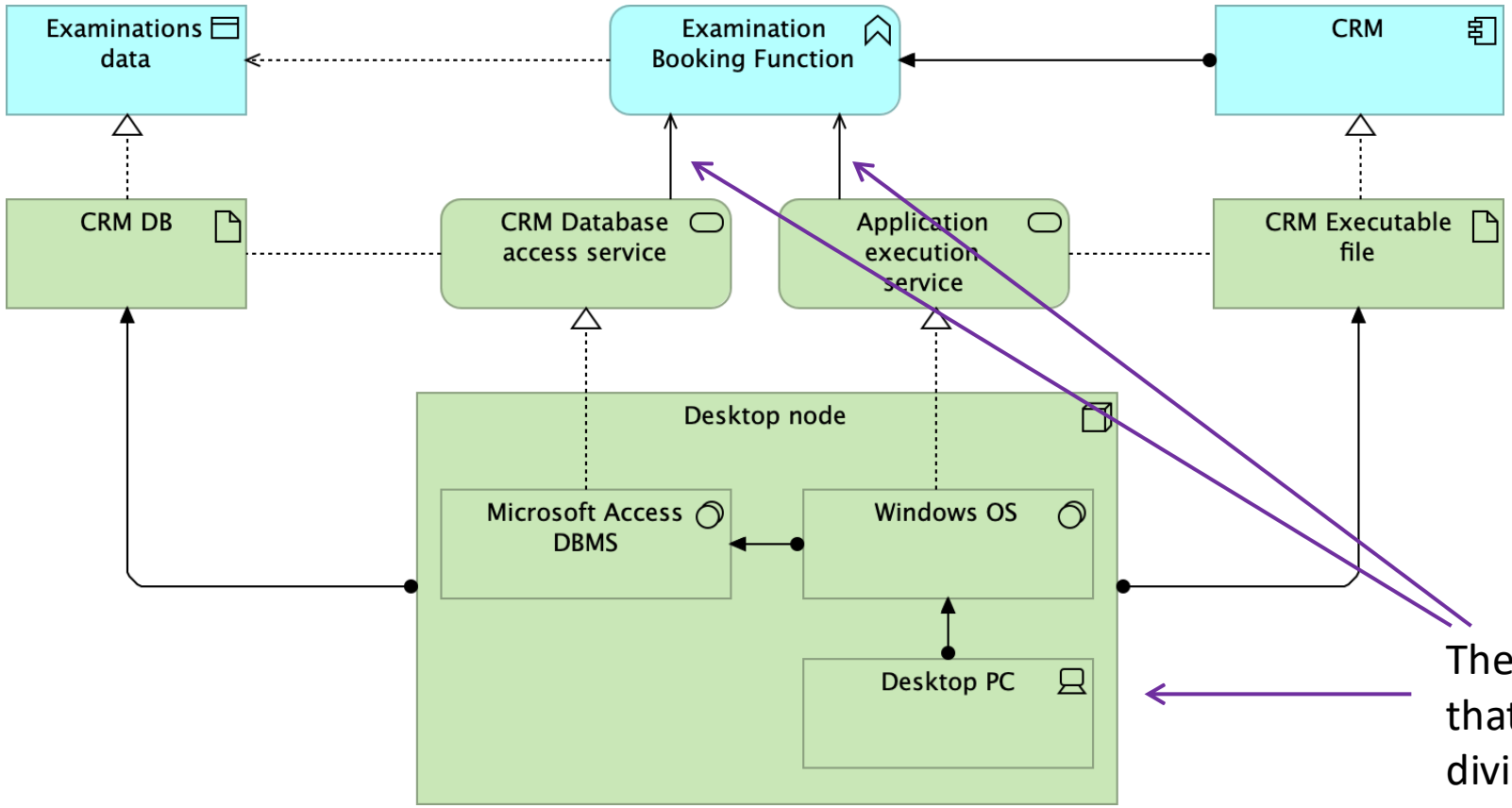
It is the only physical element (one-tiered)

One-tiered application



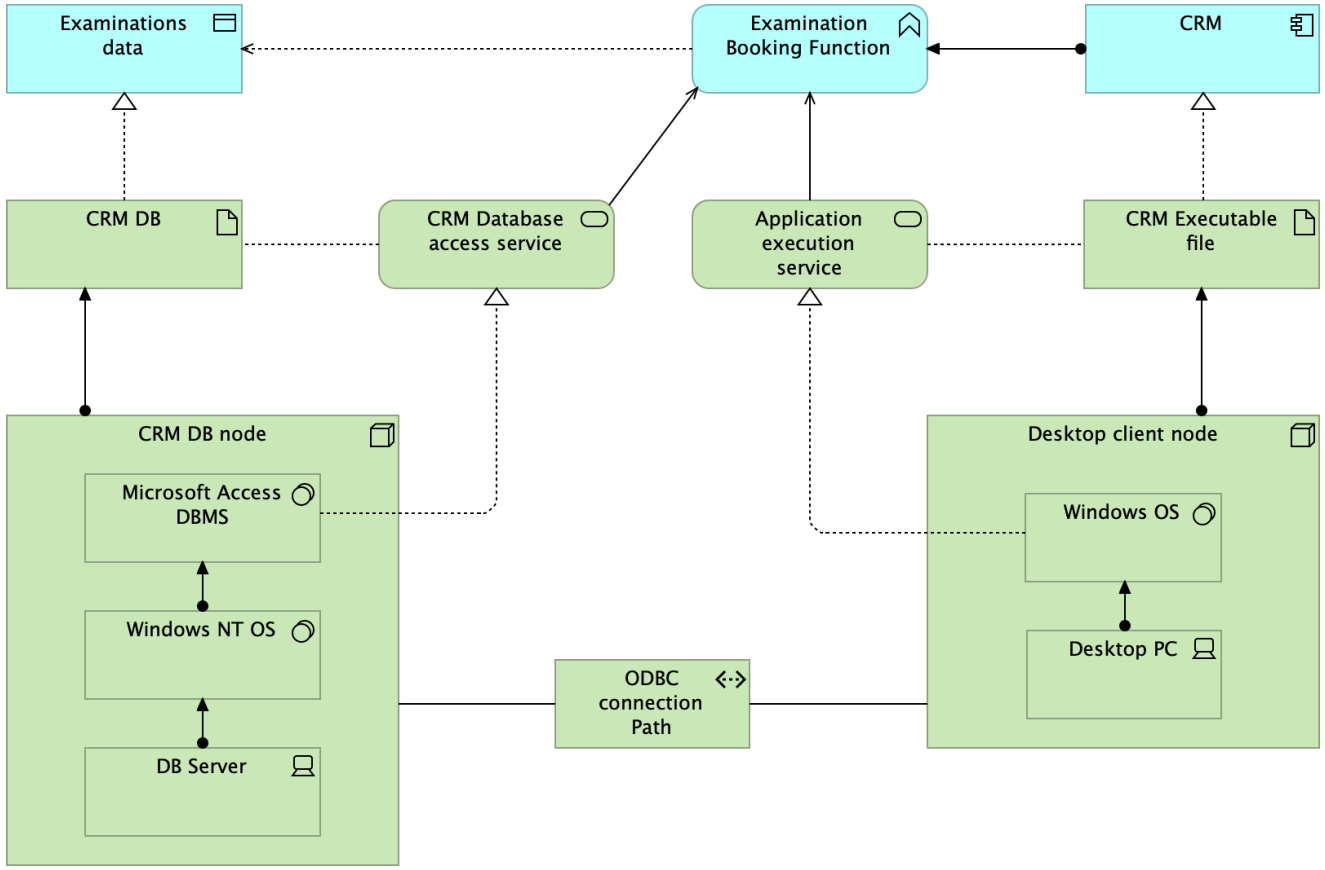
The responsibility for executing CRM.exe and DB.exe files shifts from the Device to the Node

One-tiered application

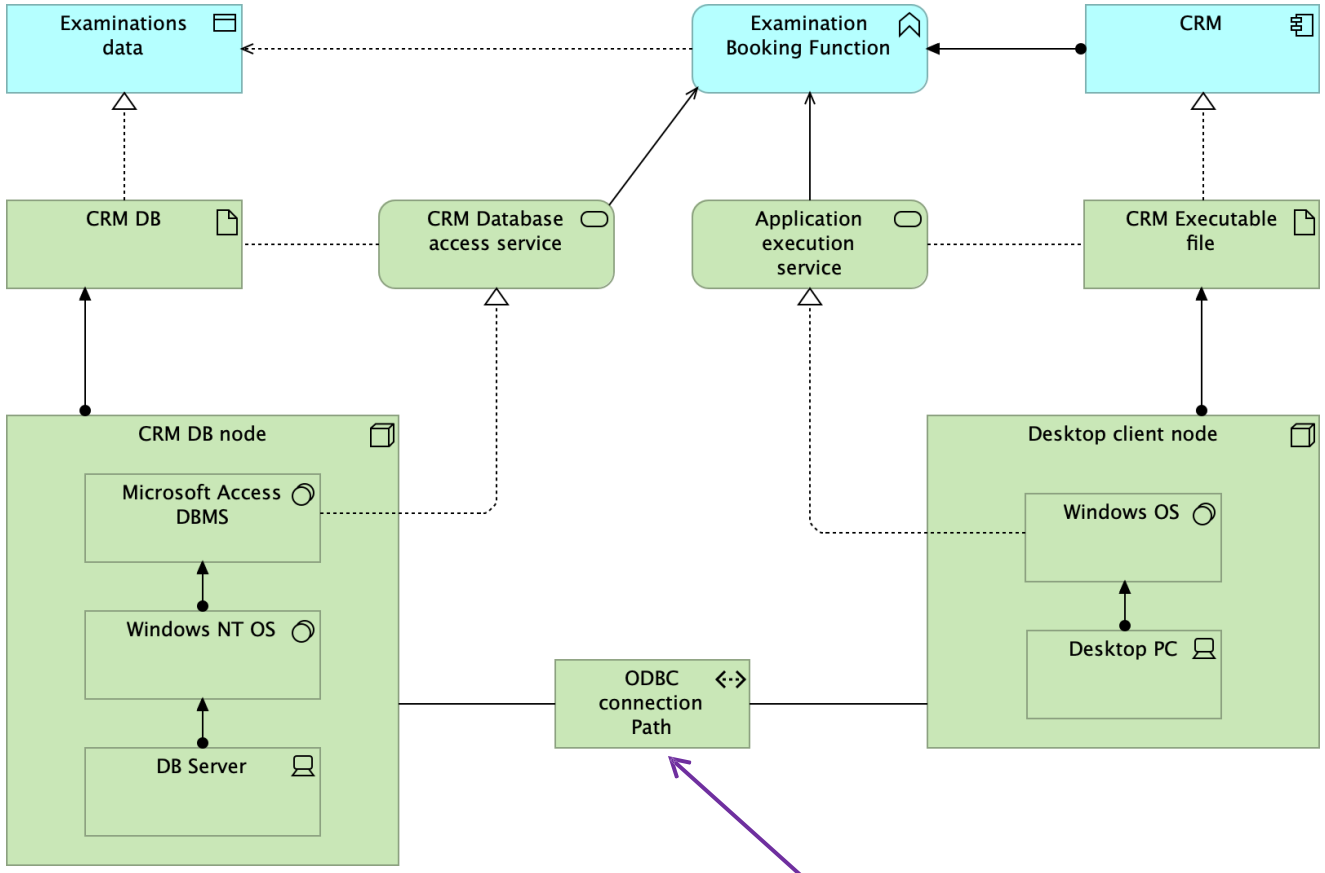


There is a single node (physical layer) that provides an application that is divided into two logical layers

Two-tiered application



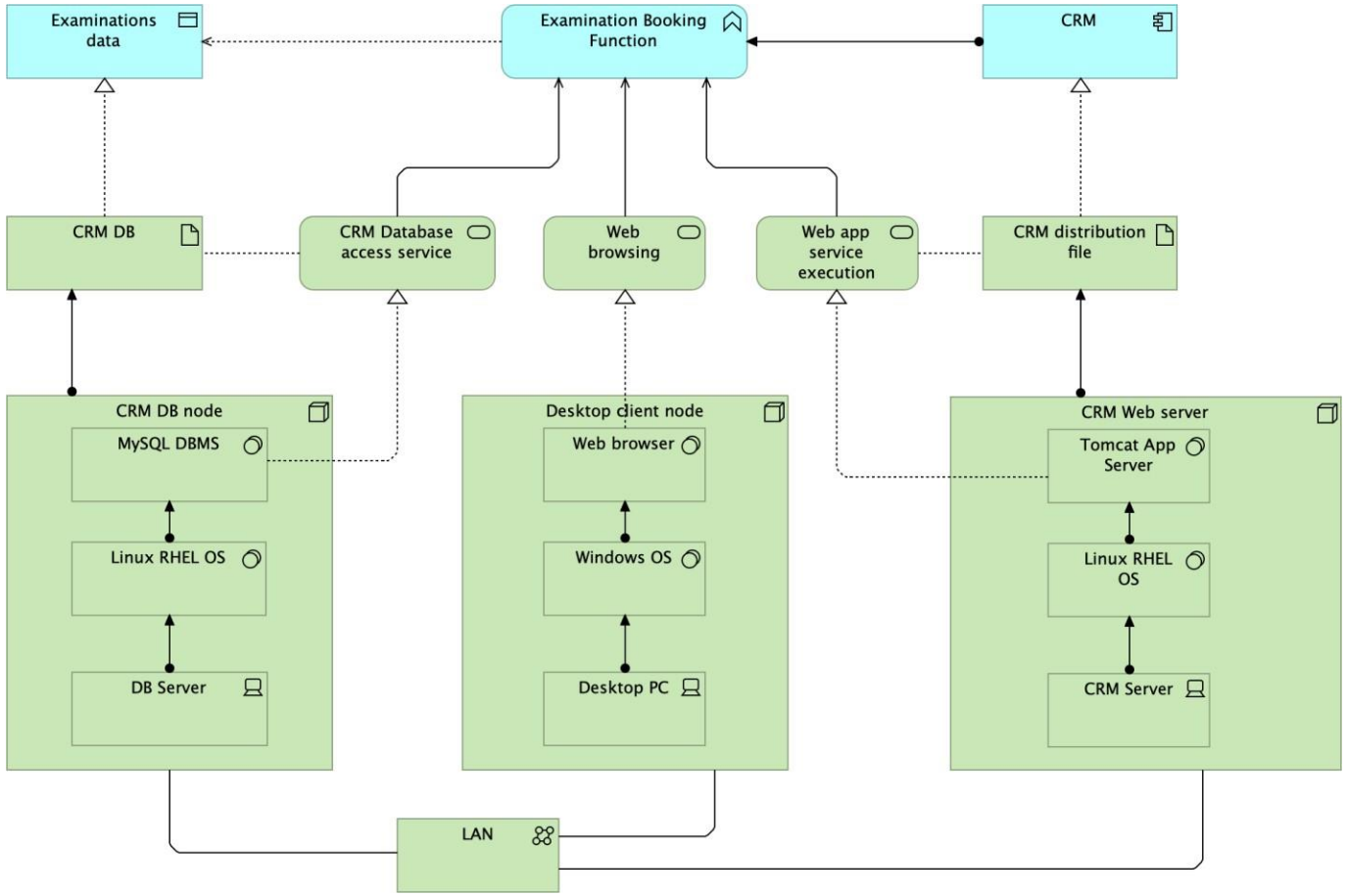
Two-tiered application



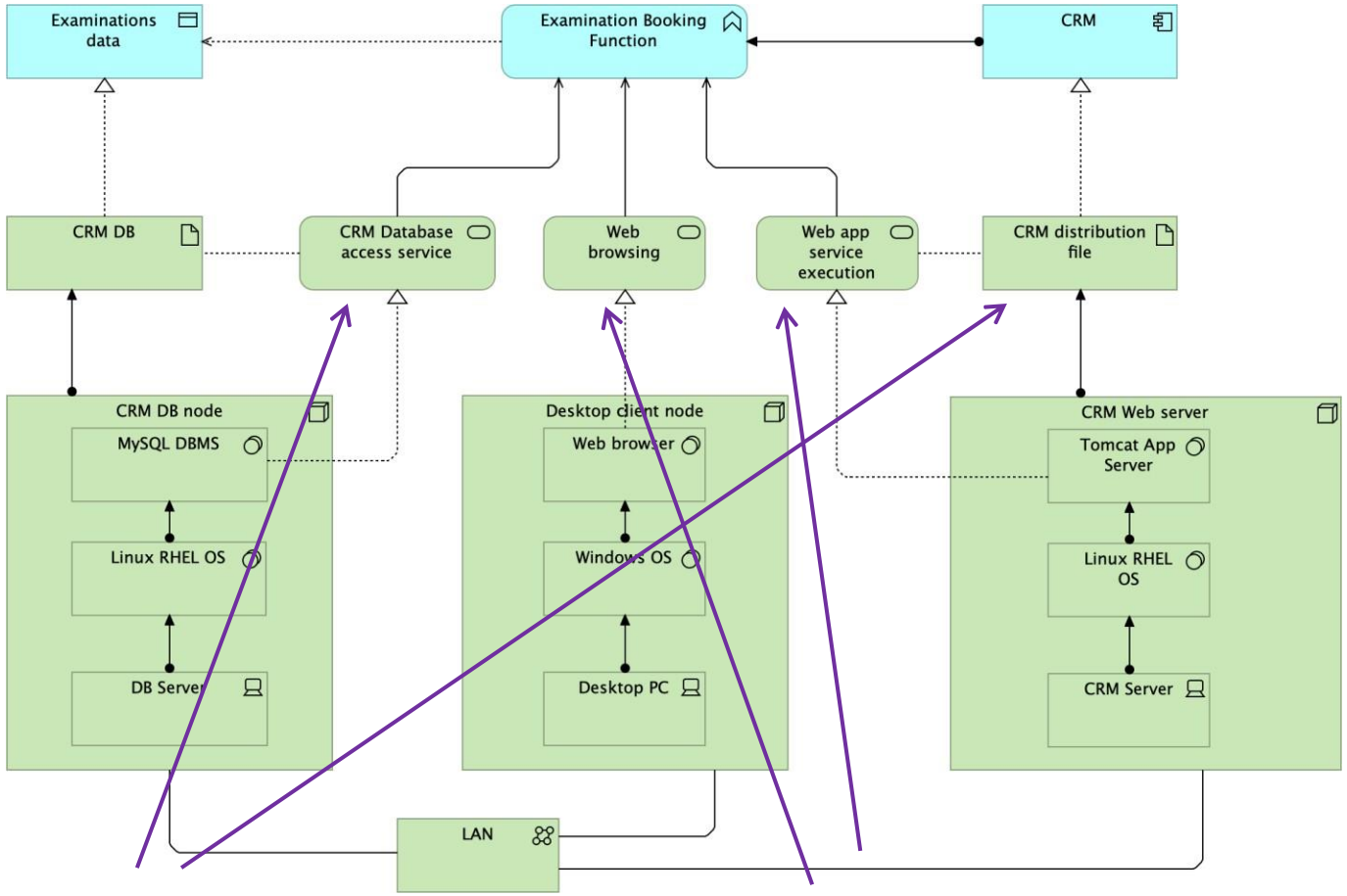
The two logical layers refer to two physical tiers (nodes)

Communication between the two nodes through a **ODBC connection** (Open Database Connectivity)

Three-tiered application



Three-tiered Application



The application offers three services:

DB / Data Access

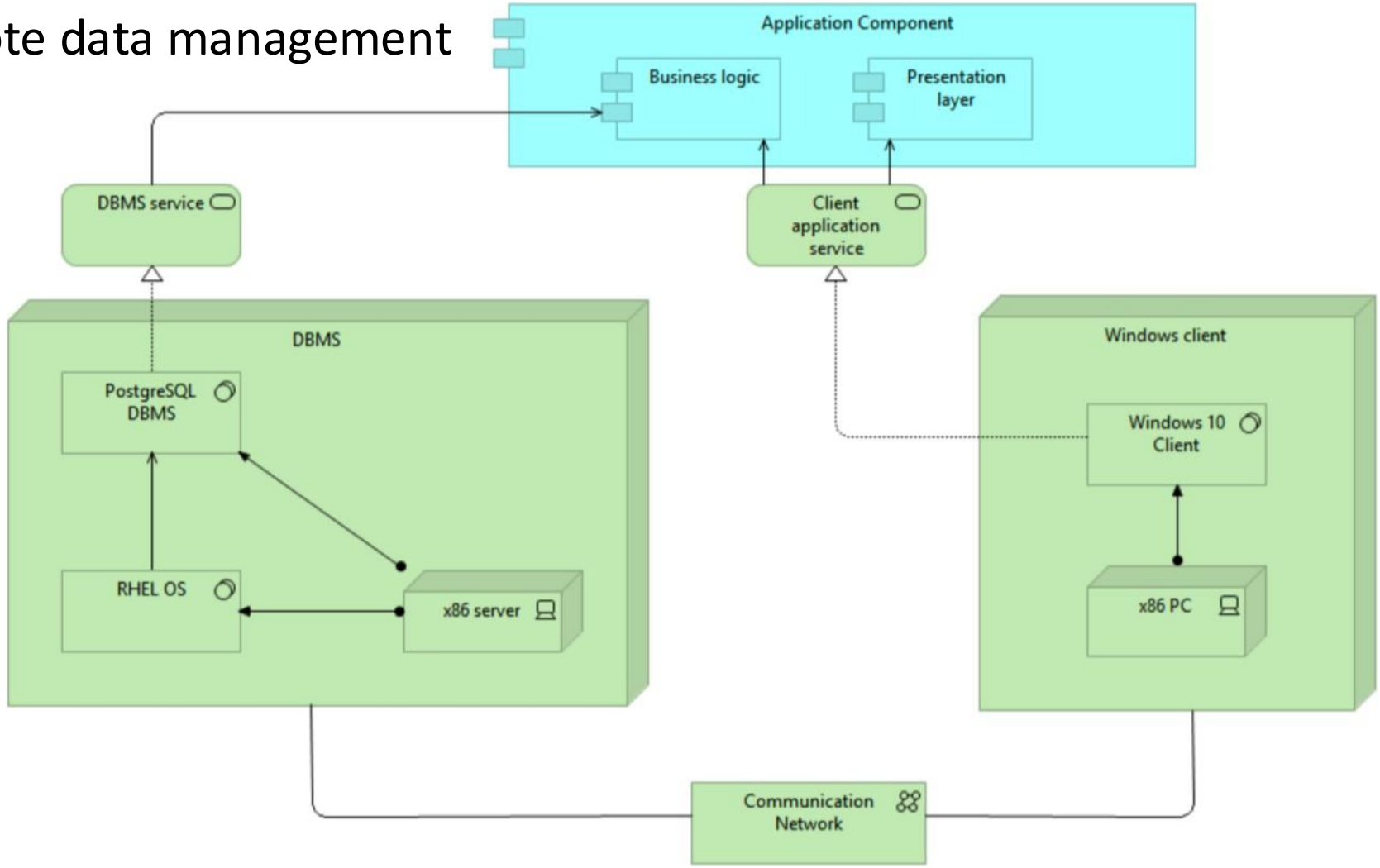
Web application

Presentation layer (browser)

The three logical layers are based on three physical nodes connected by a LAN (intranet)

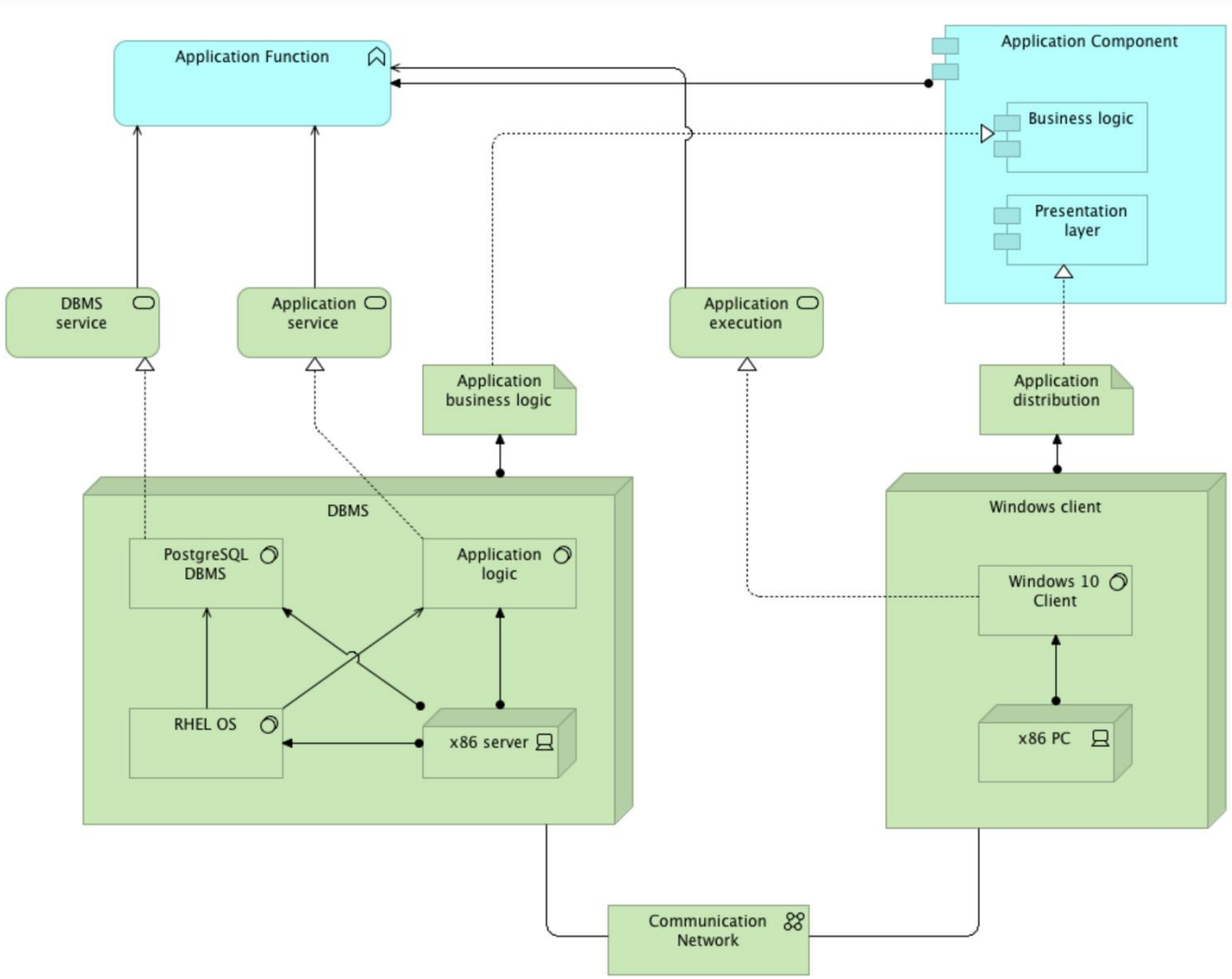
2-tier thick client

Remote data management

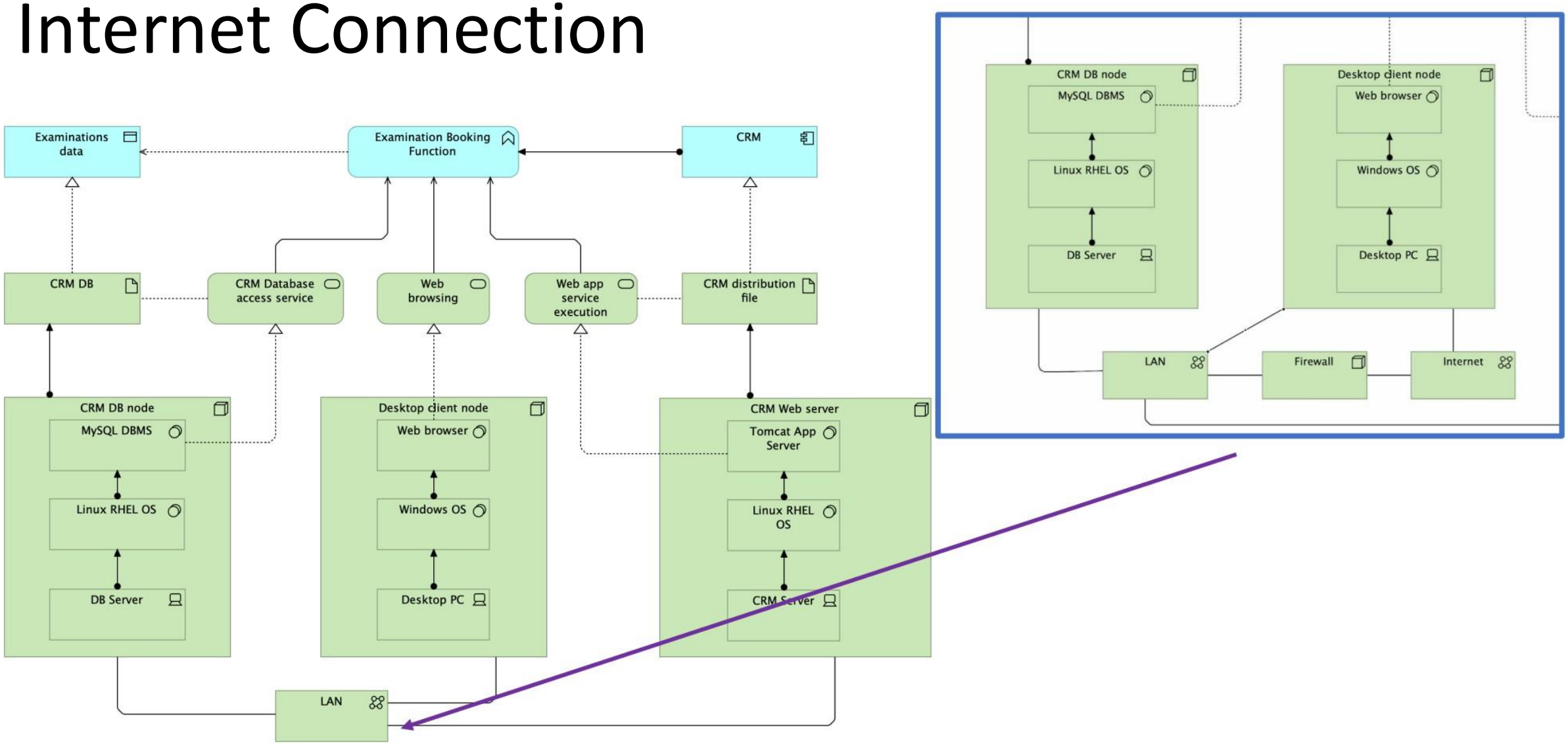


2-tier thin client

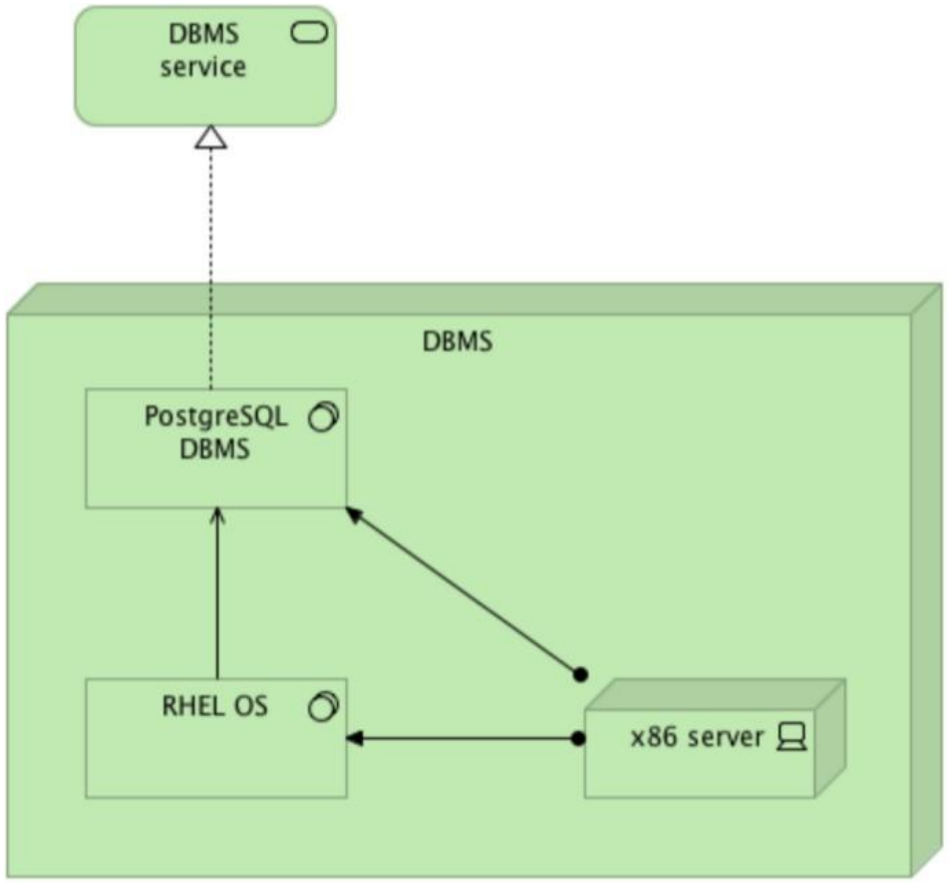
Remote Application Logic

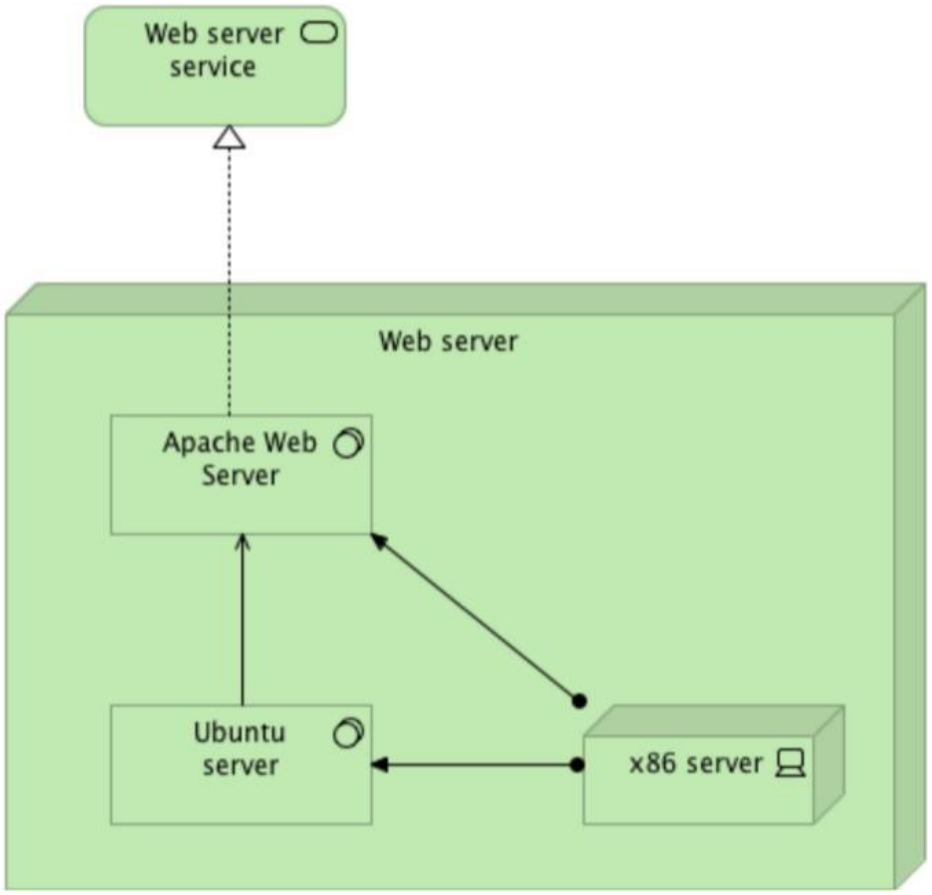


Internet Connection

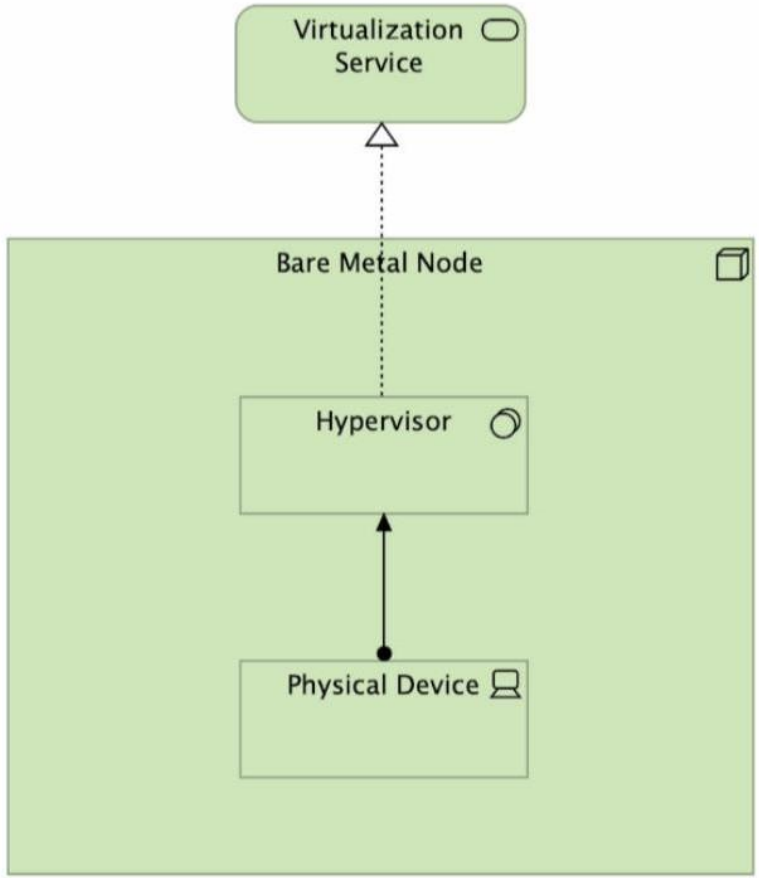


Physical server

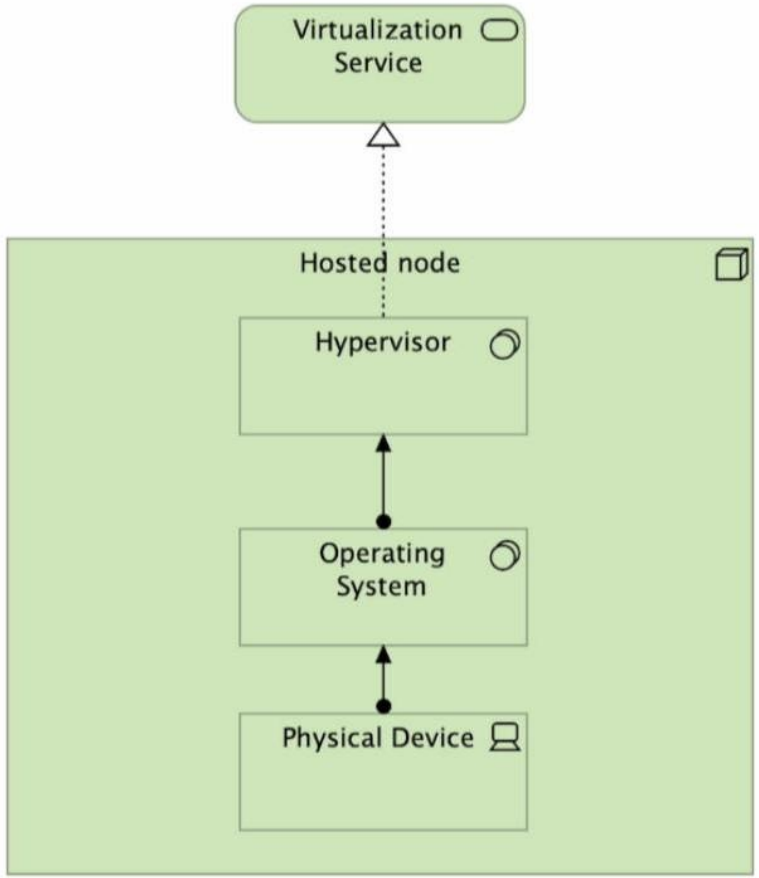




Virtualization

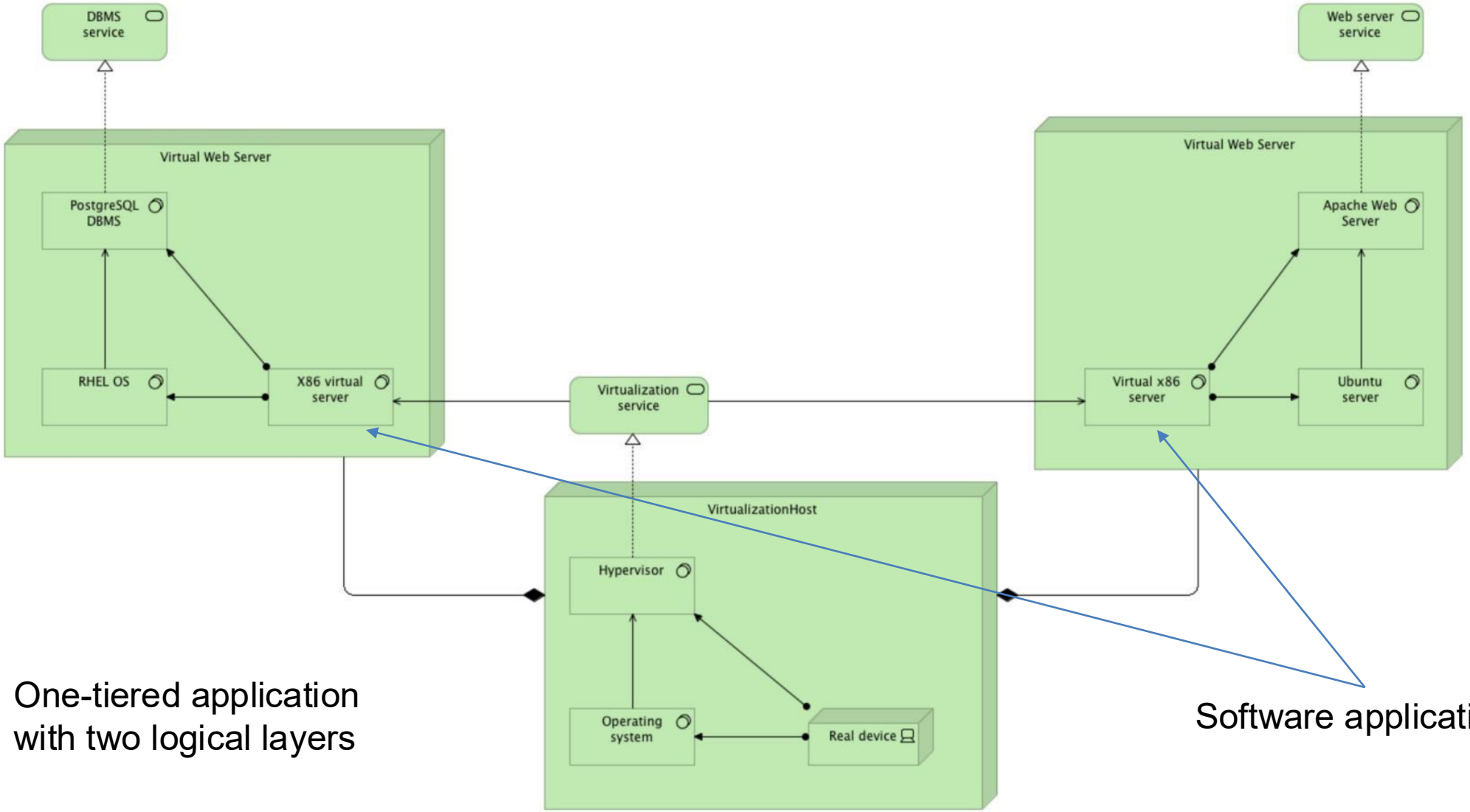


Bare – metal
(no OS)



Hosted

Virtual Servers



One-tiered application
with two logical layers

Software applications

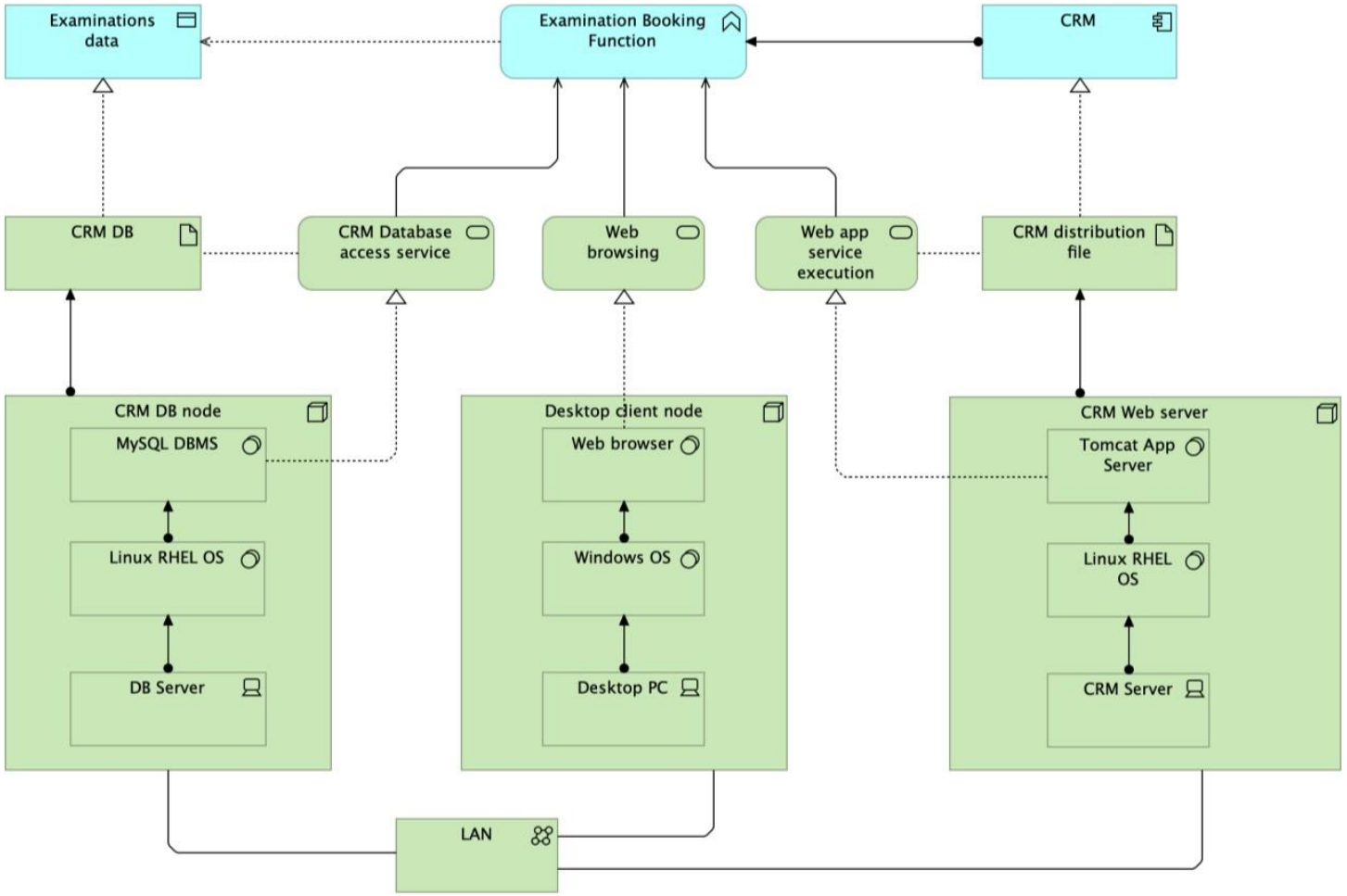
Provisioning models in cloud computing

IaaS: provides virtualized computing resources; users manage applications, data, runtime, and OS, while the provider handles the hardware

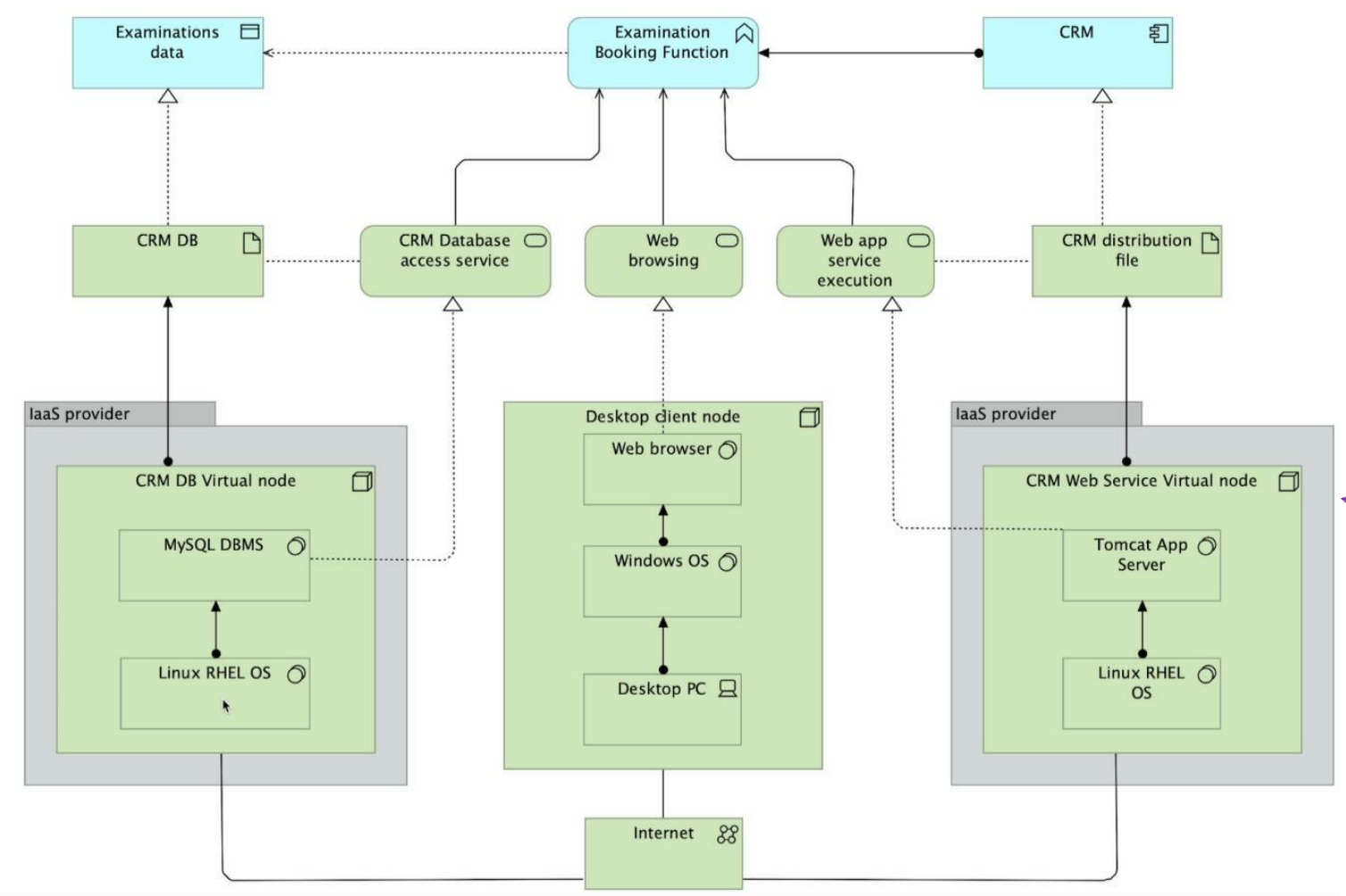
PaaS: provides managed OS, middleware, and runtime; users focus on application development, without managing infrastructure

SaaS: offers fully managed software applications over the internet

Three-tiered application



Infrastructure as a Service

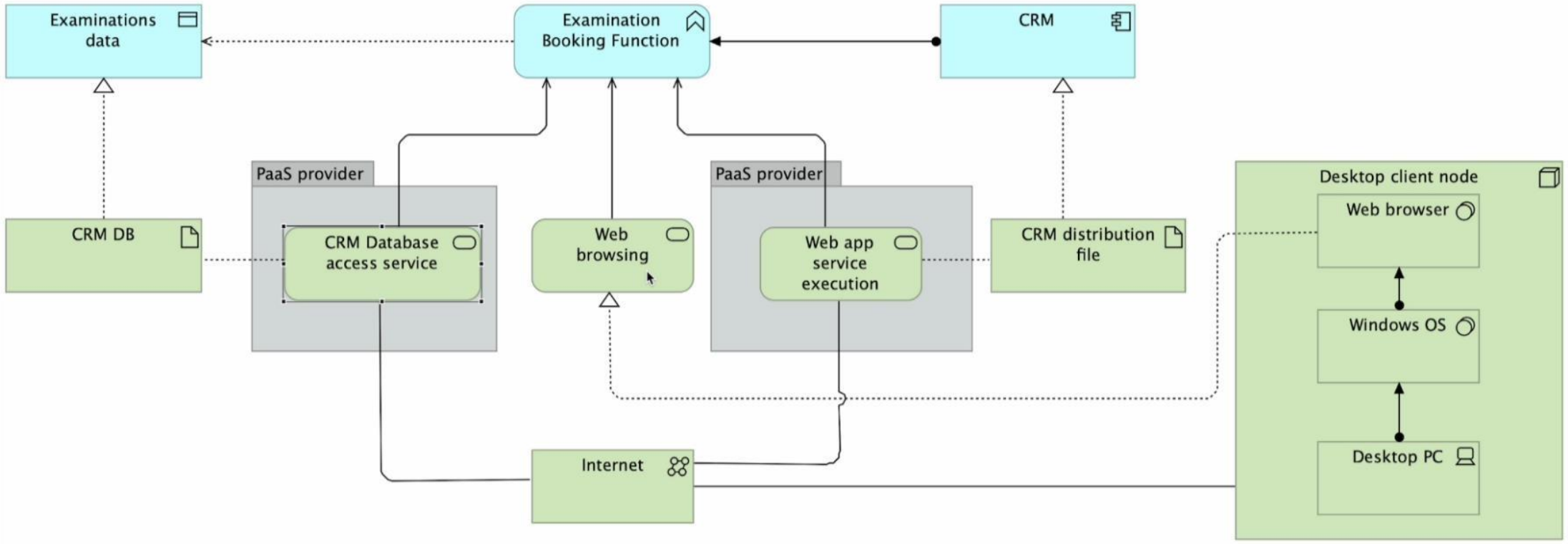


I don't know the physical devices but I only define OS

Grouping to indicate the provider offering the service

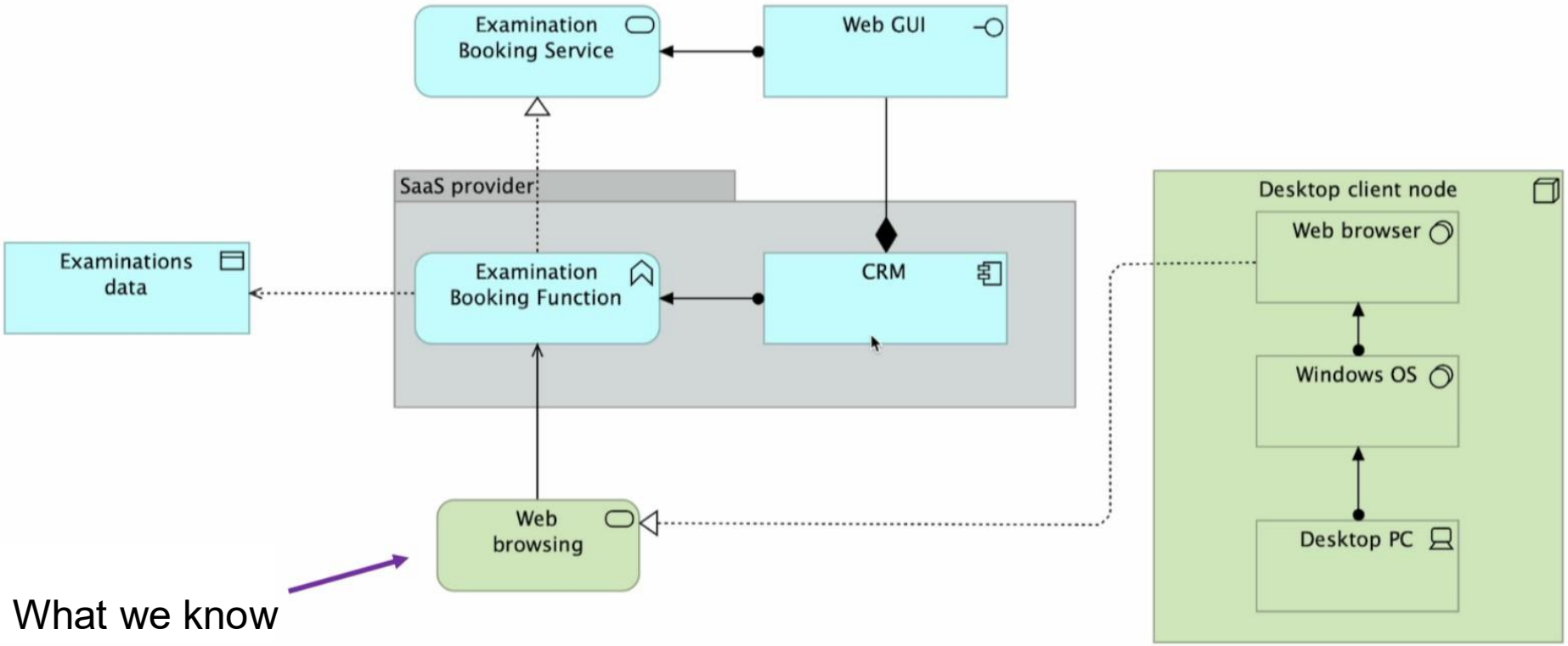
Platform as a Service

Simpler model: We don't know the underlying platform details of the services, we only represent the services



Software as a Service

Since the service is SaaS services are collapsed at the application layer. At the technological level, only the client is detailed

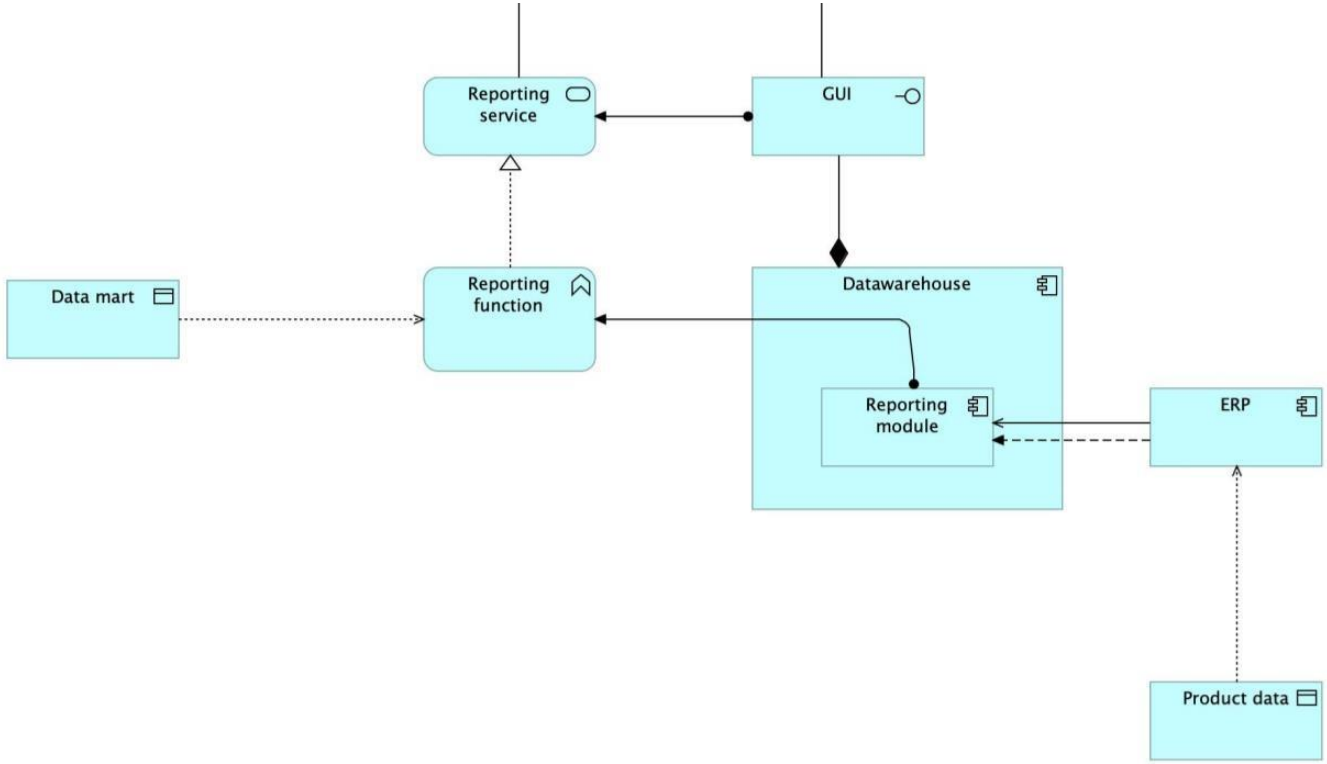


Archimate Exercises

Archi

- Speedy 01 - Application layer

To offer this service, Speedy relies on a data warehouse.
Specifically, a reporting module, which is offered by the data warehouse, is in charge of interacting with the managed data marts to enable the activities which can be performed by the user. To this aim, the data warehouse periodically reads the data about the products from the organization's ERP system through an APl, to create the data mart that is used by the reporting module.



Archi

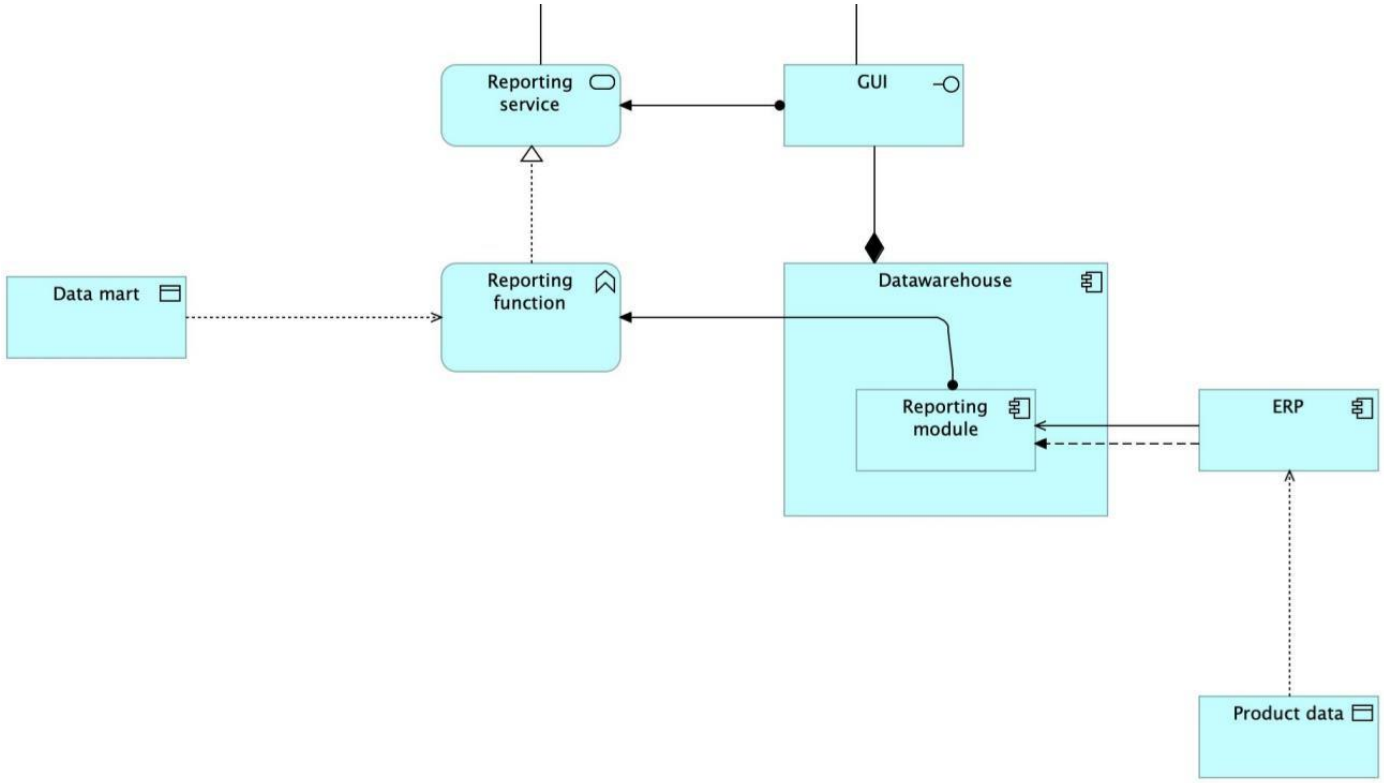
- Speedy 01 - Technology layer

From a technological perspective, the data which constitutes the data warehouse are stored in a dedicated node. This node is accessed by a data warehouse platform (that is in charge of implementing the ETL process) and also offers the possibility to navigate among the data marts and to generate the report. This platform is hosted on a server on which a Windows 2022 server instance is running. The ERP that offers the data about the product is a SaaS solution offered by the ACME cloud provider.

Archi

- Speedy 01 - Technology layer

From a technological perspective, the data which constitutes the data warehouse are stored in a dedicated node. This node is accessed by a data warehouse platform (that is in charge of implementing the ETL process) and also offers the possibility to navigate among the data marts and to generate the report. This platform is hosted on a server on which a Windows 2022 server instance is running. The ERP that offers the data about the product is a SaaS solution offered by the ACME cloud provider.



AutoMI

AutoMI is a chain of multi-brand auto body shops that intends to offer its customers a new online service for managing post-sales activities (e.g., inspections, servicing). This service allows AutoMI to send notifications to inform the customer about the need to perform periodic inspections required by law, and also allows the customer to book an appointment for servicing when a certain mileage is reached.

To provide this service, AutoMI will adopt a new software module that will interact via API, both in read and write mode, with AutoMI's internal CRM, and in read-only mode with the so-called "Portale dell'Automobilista" (Driver's Portal) made available via cloud by the Ministry of Transportation. It should be noted that the information obtained from the Portale dell'Automobilista is used only for notifications.

From a technological standpoint, the new system will be a three-tiered system deployed on virtual machines running on external cloud providers.

With reference to the previous text, define the Archimate diagram consisting of the Business Layer, Application Layer, and Technology Layer. The modeling of data objects and the technological level for AutoMI's CRM is not required.